

Improved Feature Selection using Equal Grouping Method for EEG Motor Imagery

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Abstract—Motor Imagery (MI)-based Brain-Computer Interface (BCI) systems utilize electroencephalogram (EEG) signals to interpret imagined limb movements. Due to the non-stationary and high-dimensional characteristics of EEG data, effective feature selection is essential to improve classification accuracy and reduce computational demands. This study proposes an improved feature selection method that enhances the traditional maximum relevance minimum redundancy (mRMR) algorithm by integrating the Equal Grouping Method (EGM) and Pearson correlation coefficient. EEG data from the BCI Competition IV Dataset 2b was analyzed, focusing on left- and right-hand MI tasks from nine subjects. Feature extraction employed Filter Bank Common Spatial Pattern (FBCSP) over 17 overlapping sub bands (4–40 Hz), yielding 102 features per trial. Three classification strategies were compared using Support Vector Machine (SVM): (1) raw features, (2) traditional mRMR, and (3) the proposed improved mRMR (ImRMR). ImRMR achieved the highest average classification accuracy (80.7%), outperforming mRMR (74.0%) and raw features (70.5%). It also exhibited lower variance and higher consistency across folds. The ImRMR method effectively balances relevance and redundancy, enhancing EEG-based MI classification. This hybrid approach is promising for feature optimization in real-time BCI applications.

Keywords—BCI, motor imagery, FBCSP, feature selection, mRMR

I. INTRODUCTION

Currently, electroencephalography (EEG) signals, rich in physiological and pathological information, are garnering increasing interest owing to rapid advancements in cognitive neuroscience [1]. In recent years, Brain-Computer Interface (BCI) has garnered significant interest from researchers. EEG signals are non-stationary random signals without ergodic states and are characterized by significant background noise, rendering their analysis and processing a compelling yet challenging issue [2].

Motor imagery (MI)-based BCI is a fundamental idea of brain-computer interfaces, predicated on the recording and classification of specific and temporary EEG alterations linked to the visualization of various movement types [3]. Motor imagery can alter neuronal activity in the main sensorimotor regions in a manner similar to that observed

during actual movement execution, thereby facilitating self-induced alterations in EEG. Indeed, both imagined and actual movements (or motor execution, ME) exhibit functionally analogous neuronal activation in the event-related EEG of the Mu rhythm (8–12 Hz) inside the motor cortex [4].

In classification learning, high-quality training samples are essential for the classifier development. The inclusion of irrelevant or redundant information in the dataset increases the complexity of the classification process, extends both training and prediction times, and can negatively affect the overall classification accuracy [5]. The objective is to streamline the data structure, elucidate the data of information, and the model's performance, stability, and identification of the model. Consequently, it is essential to identify the ideal features that most accurately describe an object's characteristics while eliminating irrelevant and redundant features to enhance the classifier effectiveness [6]. Feature selection is the procedure for identifying the appropriate subset of features from the original feature set to diminish the dimensionality of the feature space. The maximum relevance and minimum redundancy (mRMR) can be define as a filtering measure for feature selection in pattern recognition, designed to identify a subset of features that maximizes dependency and relevance while minimizing redundancy from the original feature set [7].

Maldonado proposed an embedded feature selection method that applies penalties to the feature set cardinality via scaling factors. This methodology was implemented in two SVM formulations specifically tailored for class-imbalanced issues, resulting in an enhanced predictive performance relative to conventional feature selection techniques [8]. Subsequently, Insik presented an enhanced mRMR feature selection technique utilizing Pearson's correlation coefficient to assess redundancy and the R-value to gauge importance. This upgrade exhibited superior classification accuracy and decreased the computational time across many high-dimensional datasets relative to the original mRMR technique [9]. Ahmed et al. proposed an improved mRMR (ImRMR) filtering technique to eliminate noisy features in ransomware malware and identified pertinent feature subset to accurately represent the true behavior of malware. ImRMR requires minimal evaluation, circumventing the superfluous

computation characteristics of the original mRMR [10]. Wang et al. examine various greedy feature selection techniques grounded in mutual information and introduced a novel class-dependent redundancy metric for feature selection. The new measure can precisely evaluate the correlation between features while considering the class variable [11].

This study makes significant contributions to the field of EEG analysis, particularly in motor imagery tasks, by focusing on improving feature selection to enhance classification accuracy and data processing efficiency. The primary contribution of this work is the development of the Equal Grouping Method (EGM), a novel approach that helps reduce feature redundancy by grouping similar features based on their information distribution. This method ensures that only the most relevant features are selected, thus optimizing the feature subset, improving the efficiency of data handling, and speeding up the classification process without compromising the quality of the results. Another key contribution is the application of the Pearson correlation coefficient to assess and manage feature redundancy. By measuring the correlation between features, this study able identify and remove highly correlated features that do not add significant value to the classification task. This helps in simplifying the model, improving its speed, and preventing overfitting, while still maintaining high classification accuracy.

This study employs an advanced machine learning classification technique to categorize features, following feature selection on the datasets by enhancing mRMR through the EGM approach as a feature subset and integrating Pearson correlation. To evaluate the effectiveness of the proposed method, conducted a comprehensive testing and assessed it using several performance metrics for accuracy. The experiment indicate proposed improved feature selection method using the SVM classifier outperformed the traditional or conventional mRMR.

II. MATERIALS AND METHODS

A. Dataset

The experimental analysis in this study employed data from the BCI Competition IV, specifically dataset IIB [12]. EEG recordings were collected from nine participants using three channels C3, Cz, and C4 at a sampling rate of 250 Hz during motor imagery tasks involving either the left or right hand. Each subject contributed 160 trials, evenly divided between the two motor imagery classes. During each session, participants received a visual cue and were instructed to imagine a hand movement for a duration of 4.5 seconds. To extract features, the raw EEG signals were first processed using a bandpass filter bank composed of overlapping frequency bands. Seventeen subbands ranging from 4 Hz to 40 Hz were applied, with each band having a 4 Hz width and a 2 Hz overlap, enabling multi-band Common Spatial Pattern (CSP) feature extraction.

B. System Flow

The initial phase of the system design delineate the system model and formulation scenarios for process analysis. Datasets frequently exhibit high dimensionality and encompass noise, redundant information, and irrelevant features that may impede the efficacy of machine learning models. Sub-band filtering is a method employed to partition raw data into distinct frequency bands, each embodying a

particular signal attribute. Dividing the data into overlapping sub-bands of 2 Hz from 4 to 40 Hz facilitates the identification of unique patterns and features pertinent to the classification task [13]. The system flow of the proposed method is illustrated in Fig. 1.

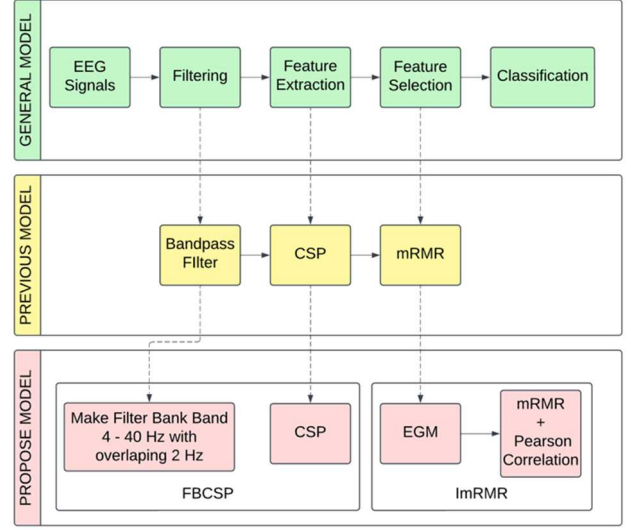


Fig. 1. System design flow.

Features from each sub band were extracted and combined to create a composite feature vector. Utilized equal grouping method (EGM) to distribute partition the candidate feature subsets. The contribution of each potential feature subset is assessed using correlation and redundancy based on mutual information and the Pearson correlation coefficient. Ultimately, the algorithm will transform ranking of the feature subsets into a comprehensive feature population ranking. Mapping the optimum feature subset to the raw sample set enables classification and recognition. A 10-fold cross-validation was conducted to assess the categorization accuracy.

C. MRMR Feature Selection

Feature selection is one of the primary challenges in big data. Maximum Relevance Minimum Redundancy, an inclusion determination calculation, creates a subset that retains the relevant qualities of the data while discarding extraneous features [14]. To assess the similarity between X and Y qualities, the calculation treats each inclusion as a discrete variable and utilizes common data [6].

The chosen characteristics were progressively incorporated into the selected feature set until the termination criterion was satisfied. Relevance (R) is defined as the mean mutual information between a feature and its category, and redundancy (D) is the mean mutual information across unselected features. Each feature was assigned a score utilizing the MRMR algorithm, with the assessment criterion for each feature computed as delineated in Eq (1) to (3).

$$\max R(F, C), \text{Relevance} = \frac{1}{|F|} \sum_{f_r \in F} I(f_r, c) \quad (1)$$

$$\max R(F, C), \text{Redundancy} = \frac{1}{|F|^2} \sum_{f_r \in F} I(f_r, f_o) \quad (2)$$

$$\max \phi(R, D), \phi = R - D \quad (3)$$

D. Equal Grouping Method

The EGM initially aims to partition the candidate group feature subsets. As shown in Fig. 2, the iterative process of assignment and shuffling directions was continued until all characteristics were allocated to the groups based on the relevance determined by mutual information calculations. In a sample set denoted as $N \times M$, N represents the number of samples, and M denotes the feature dimension. F_s denotes the feature subset derived from the EGM. This research assumed the utilization of 102 raw features, subsequently aiming to group them into $F_s = 5$, resulting in nF 20 features per group. The calculation of the member features is defined in Eq (4).

$$nF = \frac{M}{F_s} \quad (4)$$

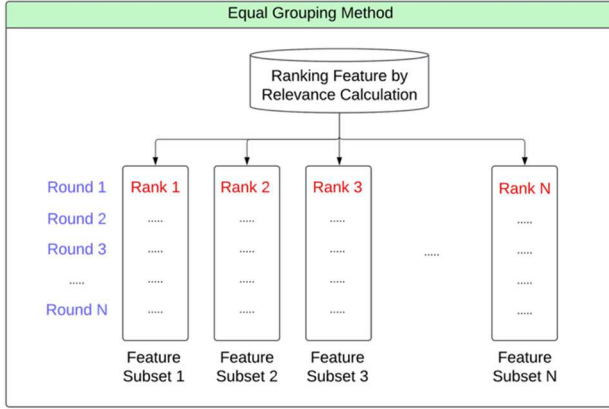


Fig. 2. Equal grouping method process.

Additionally, if 20 characteristics are not optimal for the results, the number of members should be determined for the best outcome, however the subgroups of features will diminish. Method for determining the optimal nF will employ a robustness experiment encompassing all potential nF values. It is feasible from one to half of the raw feature.

E. Improved MRMR

Pearson's correlation coefficient was a redundancy and relevance metric for the selection of an ideal feature subset. The Pearson correlation is defined in Eq (5) to (8).

$$\text{corr}(f_i; c) = \frac{\sum f_i c - \frac{\sum f_i \sum c}{N}}{\sqrt{\left(\sum f_i^2 - \frac{(\sum f_i)^2}{N}\right) \left(\sum c^2 - \frac{(\sum c)^2}{N}\right)}} \quad (5)$$

This Eq can be used for the calculation between two features f_i and f_j . This can obtain to get minimum redundancy value.

$$iD(S, c) =$$

$$\frac{1}{|S|} \sum_{f_i \in S} \beta * I(f_i; c) - (1 - \beta) * \text{corr}(f_i; c) \quad (6)$$

$$iR(S) =$$

$$\frac{1}{|S|} \sum_{f_i \in S} \beta * I(f_i; c) - (1 - \beta) * \text{corr}(f_i; c) \quad (7)$$

where $\beta \in [-1, +1]$; $iD(S, c)$ and $iR(S)$ are improved calculation of relevance and redundancy, respectively using Pearson's correlation. So ImRMR is shown in Eq (8)

$$\text{ImRMR} = \max(iD(S, c) - iR(S)) \quad (8)$$

F. Support Vector Machine

Support Vector Machine (SVM) is a machine learning technique employed to identify EEG readings through the analysis of neural activity in the brain. EEG signals were converted into a high-dimensional feature space to enable the analysis of brain activity patterns. Kernel functions are essential for SVM to effectively execute non-linear mapping, improving their capacity to identify intricate correlations within the data. Support Vector Machines (SVM) are employed to construct an ideal hyperplane that maximizes the margin for the separation of two data groups. A single hyperplane is adequate for partition two-dimensional data into two categories, that is +1 or -1 [15].

Widely utilized kernel functions encompass the Gaussian kernel, which incorporates gamma scale parameters derived from feature extraction. The Gaussian kernel equation and gamma scale are specified in Eq (9) and (10), respectively.

$$K(x, y) = \exp\left(-\frac{(x-y)^2}{2\sigma^2}\right) \quad (9)$$

$$\gamma(\text{gamma}) = 1/\text{Features} \quad (10)$$

G. Validation

K-fold cross-validation (CV) was used to mitigate variability in accuracy assessments resulting from random sampling. The K-fold Cross-Validation partitions the dataset into training and validation subsets. There are generally three criteria for selecting the number of folds, referred to as K: Representative, where $K = 10$, and $K = n$. In this study, $K = 2-10$ was selected, indicating that the dataset was partitioned into ten folds for cross-validation [16].

$$\text{CVA} = \frac{1}{k} \sum_{i=1}^k A_i \quad (11)$$

The results of this study refer to the accuracy and other classification report value as presented in Eq (12) to (15).

$$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN} \quad (12)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (13)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (14)$$

$$\text{F1 Score} = \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (15)$$

While TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

III. RESULT AND DISCUSSION

Dataset BCI Competition IV Dataset 2b, focusing on motor imagery EEG signals from subject 1 (S1). Only trials for left and right hand movements were analyzed. Each trial was extracted with a 4 second duration and a sampling rate of 250 Hz, resulting in 1000 data points per channel. However

the figures were split into 100 data points for analysis on this study.

Preprocessing includes selecting three EEG channels (C3, Cz, and C4) and three Electrooculography (EOG) channels (EOG1, EOG2, and EOG3) to observe potential eye movement artifacts. Fig. 3 and Fig. 4 shown the original EEG signals with EOG data for both the left and right hand imagery.

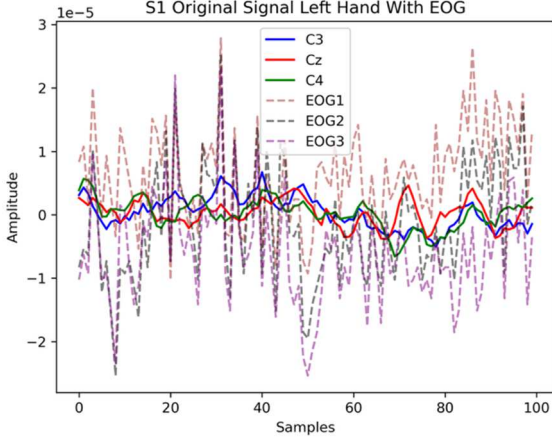


Fig. 3. EEG motor imagery left hand.

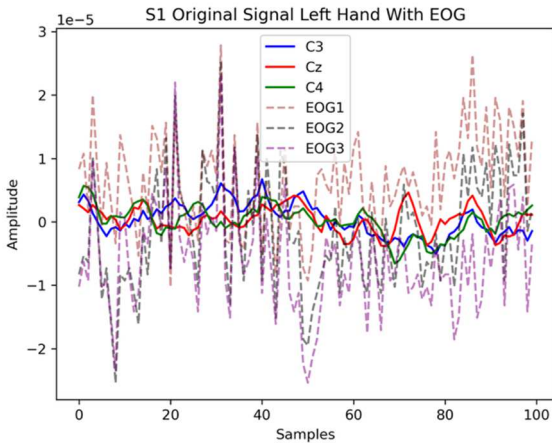


Fig. 4. EEG motor imagery right hand.

The EEG signals are centered around zero with low amplitude, whereas the EOG signals exhibit greater variability, suggesting potential noise. The addition of EOG mitigates artifacts and enhances feature quality in subsequent phases.

A. Features extraction results

This study utilized the FBSCP approach to extract discriminative features from EEG data for motor imagery categorization. The raw EEG signals were filtered into 17 overlapping frequency sub bands, each covering 4 Hz with a 2 Hz overlap, ranging from 4–8 Hz to 36–40 Hz. This overlapping structure enables the model to capture the intricate frequency dynamics pertinent to motor activities. For each sub band, six CSP components are calculated to project the EEG signals into spatial patterns that optimize the class variance between the left and right hand imagery. The features were concatenated across all sub bands, yielding a total of 102 features for each trial.

The results showed that higher-frequency sub bands (exceeding 30 Hz) have narrower distributions with overlapping quartiles and an increased number of outliers, indicating diminished importance and heightened vulnerability to noise. The existence of outliers in various sub bands, particularly in high-frequency bands, may be ascribed to residual EOG or muscle artifacts. This distribution analysis supported the hypothesis that motor-related EEG features are primarily concentrated in the alpha (8–13 Hz) and beta (13–30 Hz) bands.

B. Proposed Method

Initially, 102 features that represent various characteristics of the EEG signals. The first step involves sorting these features based on their relevance, which is evaluated using mutual information. Features will assignment and shuffling directions was continued until all characteristics were allocated to the groups based on the relevance using EGM. By parameter settings nF will 20 features per group and with 5 Feature subsets. Once the best feature subset is identified based on the Pearson correlation and redundancy reduction, this subset is selected for the next step, which is the training process. The Pearson correlation step ensures that the selected features are not highly correlated, thereby reducing redundancy and ensuring that the model only uses features with independent, meaningful information. The feature subset that results from this process is then considered the most optimal for training the classification model.

TABLE I. PARAMETER SETTINGS

Component	Description
Dataset	BCI Competition IV Dataset 2b (MI, EEG-based)
Feature Extraction	Filter Bank Common Spatial Pattern (FBCSP)
Classifier	Support Vector Machine (SVM) with RBF kernel
K Fold	K = 2-10 Fold
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, Confusion Matrix
Preprocessing	Bandpass Filtering (4–40 Hz), Normalization
Number of CSP filters	m=2, 2m=4
Number of sub-bands	17 sub bands with 2hz overlapping
Dataset	BCI Competition IV Dataset 2b (MI, EEG-based)

As shown in Table I, for the training and testing processes, a 70-30 split is used, where 70% of the data is allocated for training the model and 30% for testing. To ensure robustness and avoid overfitting, K-fold cross-validation is applied with values ranging from 2 to 10 folds.

C. Classification Result

To assess the efficacy of the suggested feature selection methods, classification was performed under three conditions: without feature selection (raw), using traditional mRMR, and employing the proposed ImRMR with equal-grouping. All scenarios were categorized utilizing SVM with the following parameters: radial basis function (RBF) kernel, C=1.0, gamma set to scale, and k-fold cross-validation (k=2-10).

In the initial situation, all 102 CSP features collected via the FBSCP approach were utilized without undergoing feature selection. According to Fig. 5 in the absence of feature selection, accuracy exhibited more variability between folds. The model intermittently fell below 70%, indicating

overfitting or interference from extraneous or superfluous characteristics within the complete 102-dimensional space.

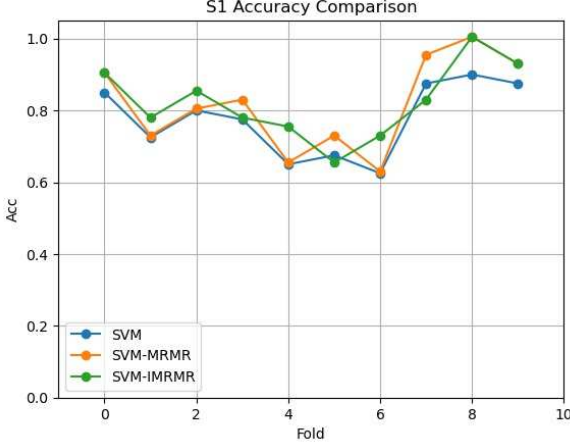


Fig. 5. Accuracy comparison.

In the second instance, mRMR, demonstrated a general enhancement compared to the raw baseline, but with somewhat more oscillations. Folds 5–7 approached 70%, suggesting that although relevance is maintained, redundancy may still influence certain trials.

The suggested technique, as illustrated, demonstrated consistently high performance in the ImRMR-based model, with the majority of folds above 85% and the optimal fold achieving 90%. The minimal variance suggests that ImRMR identifies features that generalize effectively across each fold.

Analysis of global and subject comparisons indicates that feature selection markedly improves the classification ability as shown in Fig. 6. Across all nine participants, the ImRMR technique consistently surpassed both baseline and mRMR. For instance, subject S4 achieved the most significant accuracy enhancement, increasing from 86% raw to 98% ImRMR. Similarly, ImRMR enhanced the accuracy for S3 and S6, which initially exhibited lower baseline values. The performance trend indicates that both feature selection techniques diminish noise and improve discriminability, with ImRMR exhibiting more robustness owing to its integrated relevance-redundancy assessment and group-based selection.

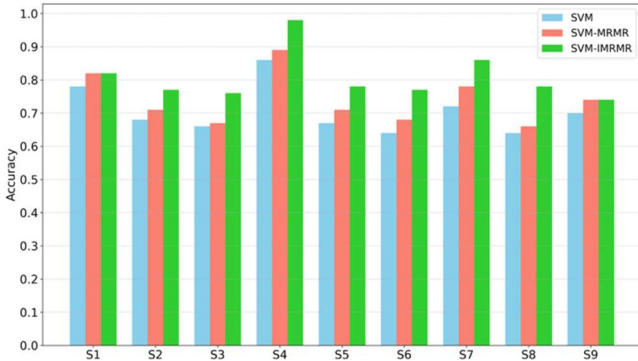


Fig. 6. Accuracy comparison method on all dataset.

Subject S1, S5, and S9 exhibit minimal discrepancies between mRMR and ImRMR, indicating that for these subject, the discriminative traits are sufficiently represented by mutual information alone. Nonetheless, for intricate or noisy datasets (e.g., S4, S6, S8), the incorporation of Pearson

correlation and the systematic selection of ImRMR yield a substantial impact.

TABLE II. ACCURACY EACH SUBJECT

Subject	Raw(Acc)	MRMR(Acc)	ImRMR(Acc)
S1	0.78	0.82	0.82
S2	0.68	0.71	0.77
S3	0.66	0.67	0.76
S4	0.86	0.89	0.98
S5	0.67	0.71	0.78
S6	0.64	0.68	0.77
S7	0.72	0.78	0.86
S8	0.64	0.66	0.78
S9	0.70	0.74	0.74
AVG	0.705	0.74	0.807

Based on Table II, the proposed method shows marginal improvements in central tendency, but it still maintains a similar range and interquartile spread, suggesting that although MRMR attempts to enhance discriminative features, it does not significantly stabilize the model performance.

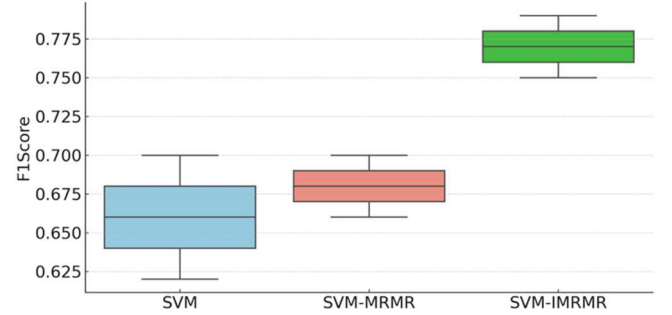


Fig. 7. F1 score comparison.

Fig. 7 provides a comparative visualization of the F1-Score distributions across three classification pipelines SVM (Raw), SVM-MRMR, and SVM-IMRMR evaluated on nine subjects under the inclusion of EOG signals and using the MIQ feature selection criterion. The raw SVM classifier exhibits relatively lower F1-Scores with a median around 0.67, and noticeable variability across subjects. Several subjects fall below 0.65, indicating inconsistency in classification performance when no feature selection is applied.

In contrast, the SVM-IMRMR classifier demonstrates a clear advancement. It not only achieves the highest median F1-Score (around 0.76–0.77), but also reflects tighter interquartile ranges, implying more consistent and reliable performance across subjects. One outlier above 0.97 indicates peak performance on certain subjects, while the overall distribution remains balanced.

IV. CONCLUSION

This study demonstrates that the proposed Improved Minimum Redundancy Maximum Relevance (ImRMR) method significantly enhances EEG classification performance by selecting a compact and informative feature subset. By integrating mutual information and Pearson correlation, ImRMR effectively identifies relevant yet non-redundant features. Compared to using raw features or

traditional mRMR, the ImRMR approach consistently outperformed in classification accuracy, achieving an average of 80.7% versus 74.0% and 70.5%, respectively. It also improved classification stability across validation folds and elevated the median F1-Score to 0.76–0.77, with reduced variability across subjects. These results confirm ImRMR's capability to reduce feature dimensionality while maintaining or improving performance in EEG-based brain-computer interface systems.

The study is limited to binary classification scenarios using offline EEG datasets. The fixed selection of 20 features and the reliance on SVM classifiers may not generalize optimally across diverse BCI applications or real-time settings. Future research should explore adaptive or dynamic feature groupings, assess the method's scalability to multi-class problems, and validate its effectiveness on clinical or real-time EEG data. Additionally, integrating ImRMR with deep learning frameworks may offer further performance gains and enhance its applicability in more complex or data-rich environments.

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